

1. What is demand paging?
- a) A paging system where pages are pre-loaded into memory.
 - b) A system where a page is loaded only when it is needed.
 - c) A method of dividing memory into fixed-size blocks.
 - d) A paging system that never requires swapping pages out of memory.

Answer: b) A system where a page is loaded only when it is needed.

2. Which of the following is a benefit of demand paging?
- a) Increases the overall memory usage of a system.
 - b) Reduces the time to load programs into memory.
 - c) Minimizes the amount of memory used by loading pages only when required.
 - d) Improves CPU scheduling decisions.

Answer: c) Minimizes the amount of memory used by loading pages only when required.

3. In demand paging, what happens when a page that is not currently in memory is accessed?
- a) A segmentation fault occurs.
 - b) The system immediately crashes.
 - c) The operating system performs a page replacement.
 - d) A page fault occurs, and the operating system loads the required page from secondary storage.

Answer: d) A page fault occurs, and the operating system loads the required page from secondary storage.

4. Which of the following algorithms is commonly used for page replacement in demand paging?
- a) First-Come, First-Served (FCFS)
 - b) Shortest Job Next (SJN)
 - c) Least Recently Used (LRU)
 - d) Round Robin (RR)

Answer: c) Least Recently Used (LRU)

5. What is thrashing in the context of demand paging?
- a) The process of repeatedly swapping pages in and out of memory, causing high overhead and slow performance.
 - b) A situation where all the pages are loaded in memory at the same time.
 - c) A memory allocation technique used to prevent page faults.
 - d) The process of dividing memory into multiple pages.

Answer: a) The process of repeatedly swapping pages in and out of memory, causing high overhead and slow performance.

6. What is Copy-on-Write (COW)?
- a) A technique where a process duplicates data immediately upon creation.
 - b) A memory management strategy where the copying of data is delayed until it is modified.

- c) A system that prevents processes from sharing memory.
- d) A technique that always results in a full copy of the memory for each process.

Answer: b) A memory management strategy where the copying of data is delayed until it is modified.

7. What is the main benefit of using Copy-on-Write?
- a) It reduces the overhead of context switching.
 - b) It minimizes memory usage by delaying the copying of shared data.
 - c) It ensures that processes have exclusive access to memory.
 - d) It speeds up the process creation by avoiding page faults.

Answer: b) It minimizes memory usage by delaying the copying of shared data.

8. In which scenario is Copy-on-Write typically used?
- a) During the creation of new threads in a multithreaded program.
 - b) During the fork system call when a new process is created.
 - c) During context switching between processes.
 - d) During swapping pages to and from the disk.

Answer: b) During the fork system call when a new process is created.

9. When does the actual copying of data occur in Copy-on-Write?
- a) When the system allocates memory to a new process.
 - b) When both processes terminate.
 - c) When one of the processes modifies the shared data.
 - d) Immediately after the fork system call.

Answer: c) When one of the processes modifies the shared data.

10. What happens if multiple processes share the same memory in a Copy-on-Write system but none of them modify the shared data?
- a) The memory is copied multiple times for each process.
 - b) The shared memory remains unchanged, and no copying occurs.
 - c) The shared memory is locked to prevent further access.
 - d) The system forcibly copies the memory for each process at regular intervals.

Answer: b) The shared memory remains unchanged, and no copying occurs.

11. Which of the following page replacement algorithms is considered to be the most optimal, but not typically implementable in practice?
- a) First-In, First-Out (FIFO)
 - b) Least Recently Used (LRU)
 - c) Optimal Page Replacement
 - d) Clock Page Replacement

Answer: c) Optimal Page Replacement

12. In the Least Recently Used (LRU) page replacement algorithm, which page is replaced?

- a) The page that has been in memory the longest.
 - b) The page that was used most recently.
 - c) The page that has been least frequently accessed.
 - d) The page that has not been used for the longest time.
- Answer: d) The page that has not been used for the longest time.

13. What is Belady's Anomaly?

- a) A phenomenon where increasing the number of frames leads to more page faults in certain page replacement algorithms.
 - b) A situation where decreasing the number of frames reduces page faults.
 - c) A problem that occurs when using the Least Recently Used (LRU) algorithm.
 - d) An anomaly in which page replacement causes a deadlock in the system.
- Answer: a) A phenomenon where increasing the number of frames leads to more page faults in certain page replacement algorithms.

14. What is the main difference between global and local page replacement policies?

- a) Global page replacement replaces pages across all processes, while local only replaces within a single process.
- b) Local page replacement replaces pages across all processes, while global only replaces within a single process.
- c) Local page replacement leads to more page faults than global replacement.
- d) Global page replacement is faster but less efficient than local replacement.

Answer: a) Global page replacement replaces pages across all processes, while local only replaces within a single process.

15. What is frame allocation in an operating system?

- a) The process of allocating memory blocks to processes.
- b) The process of assigning a set number of physical memory frames to processes.
- c) A technique to reduce page faults by caching pages in registers.
- d) A method to allocate CPU time to processes.

Answer: b) The process of assigning a set number of physical memory frames to processes.

16. Which of the following file allocation methods involves storing all blocks of a file together on contiguous blocks of disk?

- a) Linked allocation
- b) Contiguous allocation
- c) Indexed allocation
- d) Hash-based allocation

Answer: b) Contiguous allocation

17. Which of the following is a major drawback of linked allocation in file systems?

- a) It wastes a large amount of disk space due to fragmentation.
- b) It requires all file blocks to be stored contiguously.
- c) It is inefficient for direct access to a specific block of the file.
- d) It requires maintaining an index block for each file.

Answer: c) It is inefficient for direct access to a specific block of the file.

18. What is the purpose of an inode (index node) in Unix-like file systems?

- a) To store the content of a file.
- b) To keep track of the file names and directories.
- c) To store metadata about a file, such as its size, permissions, and pointers to its data blocks.
- d) To manage user access permissions for each file.

Answer: c) To store metadata about a file, such as its size, permissions, and pointers to its data blocks.

19. Which of the following components of a file system structure contains information about the files and directories?

- a) Superblock
- b) Inode table
- c) Data block
- d) Boot block

Answer: b) Inode table

20. In a hierarchical file system, what is the term used for the top-level directory?

- a) Root directory
- b) Home directory
- c) Parent directory
- d) Child directory

Answer: a) Root directory

21. What is a superblock in a file system structure?

- a) A block that stores the actual data of files.
- b) A structure that contains metadata about the file system, such as its size and status.
- c) A section of memory used for caching file system data.
- d) A special block used to store file permissions.

Answer: b) A structure that contains metadata about the file system, such as its size and status.

22. Which file system component keeps track of the free space on a storage device?

- a) Directory structure
- b) File Allocation Table (FAT)
- c) Inode list
- d) Master Boot Record (MBR)

Answer: b) File Allocation Table (FAT)

23. Which of the following is a common file operation provided by most operating systems?

- a) Process scheduling
- b) File creation
- c) Memory allocation
- d) Network communication

Answer: b) File creation

24. What does the 'open' operation do in a file system?

- a) It removes a file from the directory.
- b) It allocates space for the file on the disk.

- c) It retrieves the file descriptor and makes the file available for reading or writing.
- d) It copies the file to another location.

Answer: c) It retrieves the file descriptor and makes the file available for reading or writing.

25. Which file system operation is responsible for deleting the metadata and content of a file?

- a) Read
- b) Write
- c) Close
- d) Delete

Answer: d) Delete

26. What happens during the 'seek' operation in a file system?

- a) The system searches for a file within a directory.
- b) The current position within an open file is changed, allowing random access.
- c) The file system is reorganized to optimize storage.
- d) The file system checks for errors on the disk.

Answer: b) The current position within an open file is changed, allowing random access.

27. Which of the following is NOT a typical file system operation?

- a) Rename
- b) Move
- c) Fork
- d) Truncate

Answer: c) Fork

28. Which of the following directory structures allows a user to create the same file name in different directories?

- a) Single-level directory
- b) Two-level directory
- c) Tree-structured directory
- d) Flat directory

Answer: c) Tree-structured directory

29. What is the main advantage of a hash table-based directory implementation?

- a) Provides a hierarchical organization of files.
- b) Allows for faster file lookup times.
- c) Enables sharing of files across users.
- d) Reduces memory usage by compressing file names.

Answer: b) Allows for faster file lookup times.

30. In which directory implementation method are directory entries stored in a linked list, making it simple but slow for large directories?

- a) Linear list
- b) Tree-structured
- c) Hash table
- d) Indexed list

Answer: a) Linear list

31. Which of the following is a disadvantage of using a tree-structured directory?

- a) It does not allow file sharing.
- b) It has slow search times for locating files.
- c) It is more complex to implement than a single-level directory.
- d) It limits the number of files a directory can hold.

Answer: c) It is more complex to implement than a single-level directory.

32. What is a major drawback of contiguous allocation?

- a) It leads to external fragmentation.
- b) It requires a complex indexing system.
- c) It is inefficient for sequential access.
- d) It does not allow files to grow dynamically.

Answer: a) It leads to external fragmentation.

33. In linked allocation, each file is represented as:

- a) A set of contiguous blocks on disk.
- b) A linked list of disk blocks scattered across the disk.
- c) A list of indexed entries pointing to data blocks.
- d) A single, large block containing all data.

Answer: b) A linked list of disk blocks scattered across the disk.

34. Which of the following is an advantage of indexed allocation over linked allocation?

- a) Files can be stored in contiguous blocks.
- b) It supports direct access to file blocks.
- c) It reduces memory overhead.
- d) It avoids the need for a file allocation table.

Answer: b) It supports direct access to file blocks.

35. Which allocation method is most suitable for random access of file blocks?

- a) Contiguous allocation
- b) Linked allocation
- c) Indexed allocation
- d) Sequential allocation

Answer: c) Indexed allocation

36. Which of the following methods uses a linked list to manage free space in a file system?

- a) Bit vector
- b) Free list
- c) Grouping
- d) Counting

Answer: b) Free list

37. What is the major disadvantage of using a bit vector for free space management?

- a) It wastes a lot of space on the disk.
- b) Searching for a free block can be slow for large disks.
- c) It does not allow the reuse of free space.
- d) It requires external fragmentation.

Answer: b) Searching for a free block can be slow for large disks.

38. In the counting method of free space management, what is maintained to track free space?
- a) The total number of free blocks on the disk.
 - b) A list of all free blocks.
 - c) The address of the first free block and the number of contiguous free blocks.
 - d) A list of used blocks and the blocks that will soon be free.

Answer: c) The address of the first free block and the number of contiguous free blocks.

39. Which of the following free space management techniques is most efficient for a system that frequently allocates and deallocates large contiguous chunks of space?
- a) Bit vector
 - b) Free list
 - c) Grouping
 - d) Counting

Answer: d) Counting

40. Which disk scheduling algorithm selects the request that is closest to the current head position?
- a) First-Come, First-Served (FCFS)
 - b) Shortest Seek Time First (SSTF)
 - c) SCAN
 - d) C-SCAN

Answer: b) Shortest Seek Time First (SSTF)

41. In which disk scheduling algorithm does the disk arm move in one direction, servicing all requests until it reaches the end, then reverses direction and services requests on the way back?
- a) C-SCAN
 - b) LOOK
 - c) SCAN
 - d) FCFS

Answer: c) SCAN

42. Which disk scheduling algorithm moves the disk arm in one direction only, servicing requests until it reaches the end of the disk, then jumps back to the beginning and continues in the same direction?
- a) LOOK
 - b) C-SCAN
 - c) SSTF
 - d) FCFS

Answer: b) C-SCAN

43. What is a primary advantage of the LOOK disk scheduling algorithm over SCAN?
- a) LOOK is less complex to implement.
 - b) LOOK can reduce the average seek time because it only goes as far as the last request in each direction before reversing.
 - c) LOOK requires less disk space to store requests.

d) LOOK services requests in a more predictable order.

Answer: b) LOOK can reduce the average seek time because it only goes as far as the last request in each direction before reversing.

44. Which of the following disk scheduling algorithms processes disk requests in the order they arrive, without considering the current position of the disk arm?

a) SSTF

b) SCAN

c) FCFS

d) C-SCAN

Answer: c) FCFS